AnchorMTM: Spatially Anchored Merge Tree Maps with Topology–Position Tradeoffs

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S1 Problem Formulation (Section 3)

Sections S1–S4 follow Sections 3–6 of the paper and contain only material the paper refers to but does not include. Unprefixed equation, figure, table and section numbers refer to the paper.

S1.1 Leaf Correspondence

Supports are compared as sets of sample identifiers that denote the same spatial sample in every frame; on a fixed grid the identifier is the sample index. When overlaps suggest one-to-many, many-to-one or ambiguous relations, only the links chosen by the assignment survive (Fig. S1); split and merge events are not reconstructed, and no link bridges a missing frame. A feature whose supports in consecutive frames share no sample loses its link, so it contributes neither a temporal target nor a motion sample to Eq. 2.

S1.2 Numerical Conventions of the Direction

The signed projected motion counts as zero below 10^{-12}, and the motion energy as zero below 10^{-20}; if several domain axes are equally long, the first is used. The relative gap between the two leading eigenvalues of the motion scatter is reported with every direction, so that a nearly isotropic motion, for which the direction is ill-defined, is visible.

S1.3 Closed Form of the Fixed-Order Cost

For a fixed order write $w_k = W_{\pi(k)}$, $Q_k = q_{\pi(k)}$ and $H_k = \rho w_k/2$, and let

$$D_k = \sum_{j<k} w_j + \tfrac{w_k}{2} + (k-1)g, \qquad R_k = \sum_{j>k} w_j + \tfrac{w_k}{2} + (n-k)g$$

be the distances of the kth center from the left and right canvas ends when all intervals are packed. If $\sum_k w_k + (n-1)g \le L$,

$$\tau(\pi) = \max\Big\{0,\ \max_k (D_k - Q_k - H_k),\ \max_k (Q_k - H_k - L + R_k),\ \max_{j<k} \tfrac{1}{2}(Q_j - Q_k + D_k - D_j - H_j - H_k)\Big\}. \tag{S1}$$

Necessity. An anchor lies within H_k of a center that is at least D_k from the left end and at least R_k from the right end, and two centers in order $j<k$ are at least $D_k - D_j$ apart; each family of terms follows from one of these facts. *Sufficiency.* Write the centers as $z_{\pi(k)} = D_k + y_k$. The interval constraints become $0 \le y_1 \le \cdots \le y_n \le L - \sum_k w_k - (n-1)g$, and an anchor within distance t of its reference exists exactly when $Q_k - D_k - H_k - t \le y_k \le Q_k - D_k + H_k + t$. A nondecreasing y inside these windows exists if and only if every prefix maximum of the lower ends stays below the corresponding upper end; expanding this condition gives Eq. S1, and the prefix maxima themselves are feasible centers. The implementation evaluates Eq. S1 directly.

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S2 Proofs (Section 4 and Proposition 4)

Since equal-level merges are single nodes, levels increase strictly from every node to its parent. Leaf k lies *between* leaves i and j if its anchor does. The proofs use the separation inequality derived at the start of Section 4.3.

Lemma S1 (Contiguity)**.** *If the parent of u lies more than 2η above u, the leaves of u are contiguous in every profile with $d_{\text{top}} \le \eta$. With $\eta = 0$ this holds for every node, so an exact profile keeps every subtree contiguous.*

Proof. Take $k \notin \mathscr{D}(u)$ and $i \in \mathscr{D}(u)$. Their lowest common ancestor is a proper ancestor of u, so $V_{ik} \ge h(\text{parent}(u)) > h(u) + 2\eta$, which the separation inequality forbids if k lay between two leaves of u.

Lemma S2 (Point order)**.** *For a fixed order π, the smallest worst displacement of point anchors $x_{\pi(1)} \le \cdots \le x_{\pi(n)}$ equals $L\,\text{NROE}(\pi)$.*

Proof. For $j \le k$ with $q_{\pi(j)} > q_{\pi(k)}$, ordered anchors give $(q_{\pi(j)} - x_{\pi(j)}) + (x_{\pi(k)} - q_{\pi(k)}) \ge q_{\pi(j)} - q_{\pi(k)}$, so one of the two displacements is at least half of this inversion. Conversely, let $P_k = \max_{j\le k} q_{\pi(j)}$ and $S_k = \min_{l\ge k} q_{\pi(l)}$ and set $x_{\pi(k)} = \frac{1}{2}(P_k + S_k)$. Both sequences are nondecreasing, so the anchors are ordered, and $x_{\pi(k)} - q_{\pi(k)} = \frac{1}{2}(P_k - q_{\pi(k)}) - \frac{1}{2}(q_{\pi(k)} - S_k)$, where both brackets are inversions of the order.

Proof of Proposition 1. *Bound.* Let X, Y be children of v. By Lemma S1 they occupy disjoint contiguous blocks. If X comes first, its leaf with reference M_X precedes the leaf of Y with reference m_Y; by Lemma S2, $\text{NROE}(\pi) \ge [M_X - m_Y]^+/(2L)$. The opposite placement gives $[M_Y - m_X]^+/(2L)$, so $\text{NROE}(\pi)$ is at least the smaller value for every sibling pair, and therefore at least ν^*. *Attainment.* The midpoint order is exact because it keeps every subtree contiguous. Let leaf a precede leaf b with $q_a > q_b$, and let $X \ni a$ and $Y \ni b$ be the children of $\text{lca}(a,b)$. X precedes Y, so $m_X + M_X \le m_Y + M_Y$, that is $M_X - m_Y \le M_Y - m_X$. Hence $q_a - q_b \le M_X - m_Y = \min(M_X - m_Y, M_Y - m_X) \le 2L\nu^*$, and Lemma S2 gives $\text{NROE}(\pi) \le \nu^*$. *Time.* Hulls are merged bottom-up in $O(n)$; sorting the children of all nodes costs $O(\sum_v \deg v \log \deg v) \subseteq O(n \log n)$.

Proof of Proposition 2. Fix a layout in order π with worst displacement t; the first inequality of Eq. 10 is the definition of τ. *Two ordered sets.* Suppose all leaves of X precede all leaves of Y. Choose $S_X \subseteq X$ and $S_Y \subseteq Y$, put $S = S_X \cup S_Y$, and let a and b be the first and last leaves of S. The intervals of S lie between the centers of a and b and are separated by gaps of at least g, so $z_b - z_a \ge \sum_{k\in S} s_k - g - W_a/2 - W_b/2$; with the anchor slack, $x_b - x_a \ge \sum_{k\in S} s_k - g - W_a/2 - W_b/2 - H_a - H_b$. Averaging $t \ge q_a - x_a$ and $t \ge x_b - q_b$ gives

$$t \ge \tfrac{1}{2}\Big[\ell_a + \sum_{k\in S} s_k - h_b - g\Big] \ge \tfrac{1}{2}\Big[\min_{S_X} \ell + \sum_{S_X} s_k + \sum_{S_Y} s_k - \max_{S_Y} h - g\Big].$$

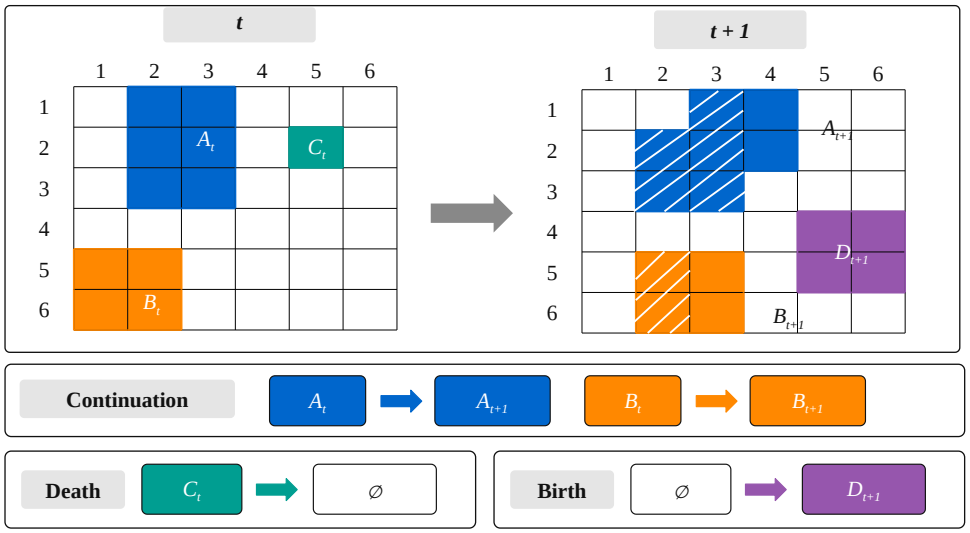


Figure S1: Leaf correspondence between frames t and $t+1$. Colored regions are leaf-arc supports, and hatching marks the samples a support shares with the previous frame. A and B continue their tracks, C_t has no successor and D_{t+1} starts a track. Birth and death are matching states, not detected physical events.

With $S_X = \{k \in X : \ell_k \geq \alpha\}$ and $S_Y = \{k \in Y : h_k \leq \zeta\}$, maximizing over α and ζ independently yields $t \geq \frac{1}{2}[A(X) + B(Y) - g]$. Sibling subtrees take one of the two placements (Lemma S1), hence $t \geq \omega(X,Y)$. *Canvas ends.* If b is the last leaf of S, the intervals of S fit in $[0, z_b + W_b/2]$, so $z_b \geq \sum_{k\in S} s_k - g - W_b/2$ and $t \geq x_b - q_b \geq \sum_{k\in S} s_k - g - h_b$. Taking $S = \{k : h_k \leq \zeta\}$ over all leaves gives $t \geq B(\mathscr{F}) - g$; the first leaf and the right end give $t \geq A(\mathscr{F}) - g - L$ in the same way. *Zero widths.* Then $\ell = h = q$ and $s = 0$, so $A(X) = M_X$, $B(Y) = -m_Y$ and $\omega(X,Y) = \frac{1}{2}\min(M_X - m_Y, M_Y - m_X)$, while both canvas terms are nonpositive because $q \in [0, L]$.

Proof of Proposition 3. Let $k \notin \mathscr{D}(A_\eta(u))$ and $i \in \mathscr{D}(u)$. Then $\mathrm{lca}(i,k)$ is a proper ancestor of $A_\eta(u)$, whose level exceeds $h(u) + 2\eta$ by the maximality of $A_\eta(u)$; by the separation inequality, k lies between no two leaves of u, which settles the pairs $(\mathscr{D}(u), \{k\})$. For a pair $(\mathscr{D}(u), \mathscr{D}(w))$, the two disjointness conditions give the same statement for every leaf of w relative to u and every leaf of u relative to w. If the earliest of these leaves belongs to u and some leaf of u followed a leaf of w, that leaf of w would lie between two leaves of u; so all leaves of u precede all leaves of w. The two possible placements of each pair then give Eq. 11 exactly as in the proofs of Propositions 1 and 2.

The flattened bound. Contract every internal node u with $h(\mathrm{parent}(u)) - h(u) \leq 2\eta$. By Lemma S1, every node that remains is contiguous whenever $d_{\mathrm{top}} \leq \eta$, so the order keeps every subtree of the contracted tree contiguous, and Propositions 1 and 2 apply to that tree. The two bounds are incomparable. Two leaves of width 0.2 with equal references under one root give an empty $\mathscr{P}_\eta$, whereas the contracted tree yields 0.1 (with $\rho = g = 0$). Conversely, a chain of ancestors whose single gaps are at most 2η but whose accumulated gap exceeds it separates pairs that the contraction merges.

Proof of Proposition 4. Let b_k be the maximum of ψ on the open interval from the kth to the $(k+1)$th anchor. The profile is continuous and both anchors are minima, so $b_k \geq F_k$, and the displayed merge level of the ith and jth leaf is $\max_{i\leq k<j} b_k$. A merge error of at most δ requires (i) $b_k \leq C_k + \delta$, because no pair spanning gap k may be displayed too high; (ii) $\max_{i\leq k<j} b_k \geq V_{\pi(i)\pi(j)} - \delta$ for every pair; and (iii) $b_k \geq F_k$. Raising barriers cannot violate (ii) or (iii), so some choice is feasible if and only if $b_k = C_k + \delta$ is. This is the case exactly when $\delta \geq F_k - C_k$ for every k and $2\delta \geq V_{\pi(i)\pi(j)} - \max_{i\leq k<j} C_k$ for every $i < j$; the least such $\delta \geq 0$ is Eq. 13. Any admissible barriers are realized by a profile with one peak per gap. For an exact order the ultrametric property makes every term of Eq. 13 nonpositive.

A direct evaluation of Eq. 13 costs $O(n^3)$ per order.

S3 Construction (Section 5)

S3.1 Candidate Generation and Refinement

There is one *threshold flattening* per distinct parent gap up to 2η; it contracts every internal node whose parent gap does not exceed that value. The *selective flattening* starts from the exact tree, finds the sibling pair with the largest order conflict, and tries to release one or both of its internal blocks while respecting accumulated ancestor gaps. Trials must pass the merge-error test and are compared by NROE, then merge error, then the number of released nodes; steps that do not increase NROE may continue the search, and the best strictly improving state is kept. The best admissible midpoint order stays in the pool, so the final choice is never worse than it in τ.

Refinement starts from the exact tree and from the four best structures, ranked by τ, then by the sum of squared positive terms of Eq. S1, then by the number of released nodes. A move takes one child block of an internal node and inserts it elsewhere among its siblings: at every position for nodes with at most eight children, otherwise within distance three. In each pass, the internal nodes are visited in traversal order and the first move that improves (τ, squared sum) and passes the merge-error test is applied; refinement stops after eight passes or when no move is accepted. Every visited order enters a deduplicated pool. Exact ties in Eq. 12 are broken by the permutation itself so that runs are reproducible.

S3.2 Placement Solver

For the selected order, τ comes from Eq. S1, and the capped QP is solved by a constrained quadratic solver followed by numerical residual checks. Every valid match enters E_M with unit weight.

S3.3 Reconstruction and Rounding

Leaf intervals. The regular samples of a leaf arc are resampled to the free pixel positions of its interval and alternate between the two sides of the anchor; an arc without regular samples is interpolated between its end values. *Gaps.* The lowest common ancestor of two consecutive leaves decides which internal paths fill the gap between them; direct siblings, a direct and a deep leaf, and two deep leaves differ in how many internal paths contribute, and the

Table S1: Planar parameter settings and runtime (seconds).

Parameter	Gaussian	Ring	Wildfires	Storms	PM2.5
N_r	4096	196	4096	4096	4096
s_w	0.00581	0.00328	0.00053	0.01170	0.00073
t_{opt}	0.156	0.150	3.324	2.600	199.975
t_{total}	1.340	0.486	4.424	63.447	231.177

s_w is the width per unit of support measure and N_r the raster length; $\rho = 0.5$ throughout. Times are single runs without a tree cache: t_{opt} covers ordering and placement, t_{total} the full pipeline.

ancestor level is inserted between them. After the barrier correction, interior samples may differ from the original samples; the input field itself is unchanged. *Rounding.* Anchors and interval ends are converted to pixels by one affine map for the whole sequence. The renderer keeps every extremum strict, which can raise a barrier slightly above its continuous optimum; for this reason the merge error is measured on the rendered column rather than taken from Eq. 13. *Failures.* Widths and gaps that exceed the canvas, too few pixels for the intervals, nonfinite values and solver failures are reported for the frame and not repaired. The per-frame records keep the requested budget, the bounds, τ, δ^*, the rendered merge error and the final displacement separately.

S4 Evaluation (Section 6)

S4.1 Inputs

Because the processed arrays of the ST-MTM authors are not available, the four benchmark inputs are reconstructions: Gaussian has three peaks, Ring uses the archived generator, Wildfires aggregates fire radiative power into 32 grids, and Storms applies the archived moving mean and smoothing to the ERA5 anomalies. The ST-MTM persistence fractions $(0.1, 0.1, 15, 10.5)\%$ were checked on all 880 frames. $PM_{2.5}$ is interpolated onto a 160×144 grid. For Cylinder, Q is computed from the original $192 \times 64 \times 48$ velocities before cropping, and $-Q$ is scaled once for the whole sequence [1, 4]. For Tangaroa, speed is clipped to $[0, 2]$ and values below 10^{-6} are replaced by the inflow value 1 [3].

S4.2 Measures and Baselines

TW uses $k = (1, 3, 5, 3)$ neighbors on the four benchmark inputs and $k = 1$ on $PM_{2.5}$, with $0 < k < n_t/2$; frames with too few features are skipped. The curves pad the grid to a power of two, skip padded samples, and map the rank of each extremum among all grid samples to the canvas at the raster length of the sequence; they impose neither the tree nor finite widths [2].

ST-MTM on the volumes. The runs use the general parameters of ST-MTM without any tuning, and its geometric objective takes 3-D Euclidean distances between support centroids. Feature widths are physical volumes on Tangaroa and sample counts elsewhere, and the convex placement problem is solved with OSQP. The implementation needs binary trees, so a multifurcation is refined into binary nodes at the same level with fixed identifiers; this keeps every pairwise merge level but restricts the leaf orders that ST-MTM can choose.

S4.3 Settings

The Balanced budget minimizes $J = \frac{1}{2}\widehat{\mathrm{NPE}} + \frac{1}{2}\widehat{\mathrm{SNS}}$ over candidate budgets in 1% steps, where hats denote min–max normalization over the candidates whose attained maximum merge error stays within their budget; within 0.01 of the minimum, the smaller budget and then the smaller attained error are preferred. Budgets were fixed before the full runs. The full results include the selection frames, and exploratory runs preceded the selection, so the selection is not a held-out test. Table S1 gives the remaining planar settings and the runtimes. The volumes use the default configuration with $\rho = 0.5$, a reference direction estimated from the motion on Tangaroa and the streamwise axis on Cylinder.

S4.4 The Budget Sampled Finely

Fig. S2 repeats the sweep of Section 6.4 with finer budget steps on the three benchmark sequences that contain conflicts. Displacement falls in steps, not continuously. Between steps, extra budget is not spent: the attained merge error stays flat and below the diagonal, as on Ring from 8 to 10% and on Wildfires beyond 5%. This is consistent with Proposition 3, under which a conflict is released only once twice the budget reaches its merge-level gap. Across all budgets, SNS moves by less than 0.025 on these sequences, so moving features toward their references leaves their relative distances largely intact.

S4.5 Cylinder in Three Maps

In Fig. S3, ST-MTM spreads the shed vortices by their relative distances, so its bands drift across the axis as the wake develops.

S4.6 Hourly $PM_{2.5}$, 22–27 January 2017

High-concentration features are the leaves of the join tree of the negated field; the display restores positive values. All three maps in Fig. S4 show the build-up from 23 January and the peak on 26 January, but they place the polluted features by different rules. TMTM and ST-MTM collect them into a compact band near the middle of the axis wherever they are. In AnchorMTM the band is as wide as the polluted area along the reference direction: on 25 January (c), when pollution covers most of the domain, it spans most of the axis, and on 26 January the strongest maxima drop to the lower part of the axis while TMTM keeps them in the middle. The compact bands are therefore not a more faithful picture; they omit where the features are. The vertical stripes of the AnchorMTM map are a real cost. On this sequence nested maxima cross each other along the direction more than on any other (Section 6.2), so the budget is spent in almost every hour (Section 6.4) and the leaf order changes from hour to hour. Over all 744 hours, the map spends at most 4.998% merge error. TMTM and ST-MTM are rebuilt with their original configurations, and their anchors match the stored table runs frame by frame.

S4.7 Recomputation

Controlled trees: a separate minimax LP for each of the 24 orders, compared with all 43 stored overlap records and 364 stored budget records; for widths of 0, 3, 6 and 9%, the LP optimum equals the width-aware bound to within 10^{-16}. *Planar sequences*: per-frame displacement recomputed from the rendered anchors for all 9,168 frame–budget pairs of Ring, Storms and $PM_{2.5}$; one $PM_{2.5}$ QP at 2% failed numerically and was solved by a second solver with unchanged objective, constraints and budget, and its KKT certificate is kept with the records. *Volumes*: feature identities and source merge matrices checked against the baseline input, and every displayed pairwise merge rebuilt from the raster columns, for 1,515 frame–budget columns and 44,785 feature observations; baseline reflection is chosen by the feature-pooled absolute displacement.

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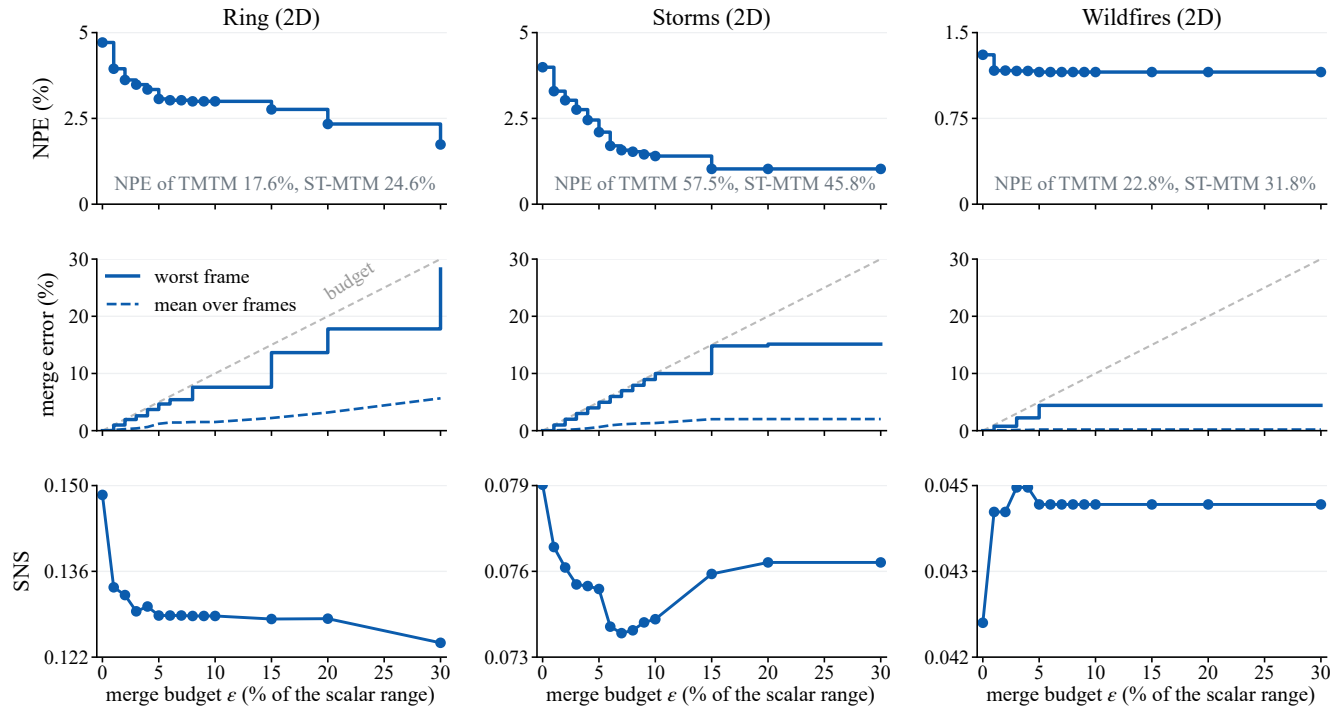


Figure S2: Ring, Storms and Wildfires at 14 budgets from 0 to 30%. (a) NPE; the grey text gives the NPE of TMTM and ST-MTM from Table 1. (b) Attained merge error of the worst frame (solid) and its frame average (dashed); the dotted diagonal is the budget. (c) SNS. TW is not drawn because these sequences have too few eligible frames (Table 1).

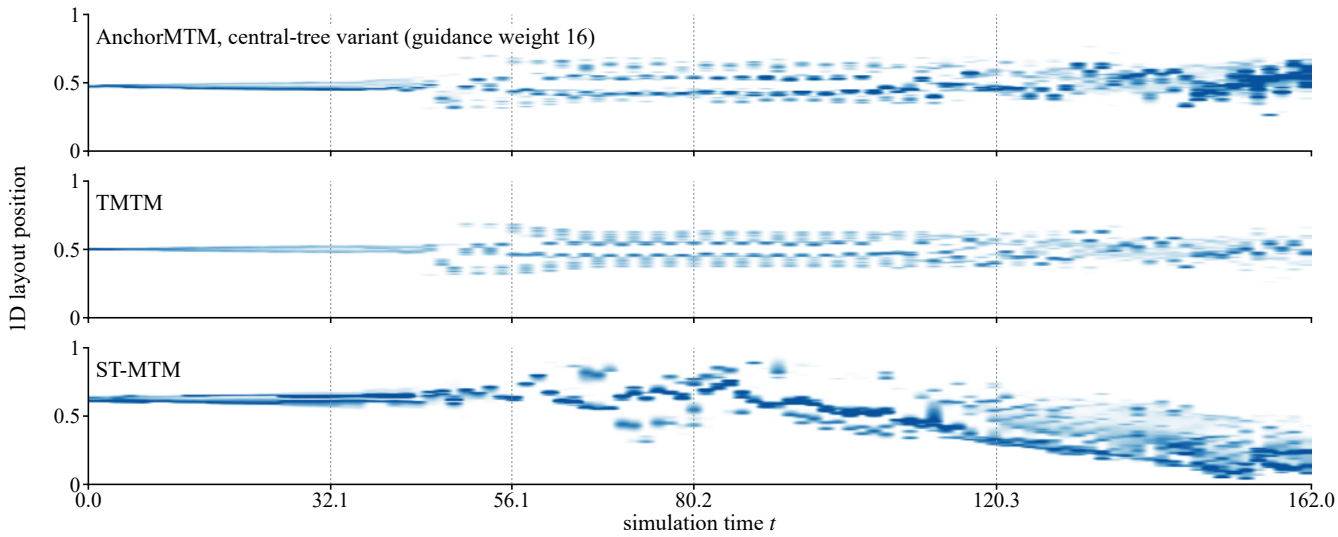


Figure S3: Cylinder, all 102 frames: the central-tree variant of AnchorMTM (guidance weight 16), TMTM and ST-MTM, all three built from identical simplified trees. Each map uses its own layout coordinate; dotted lines mark the snapshot times of Fig. 10.

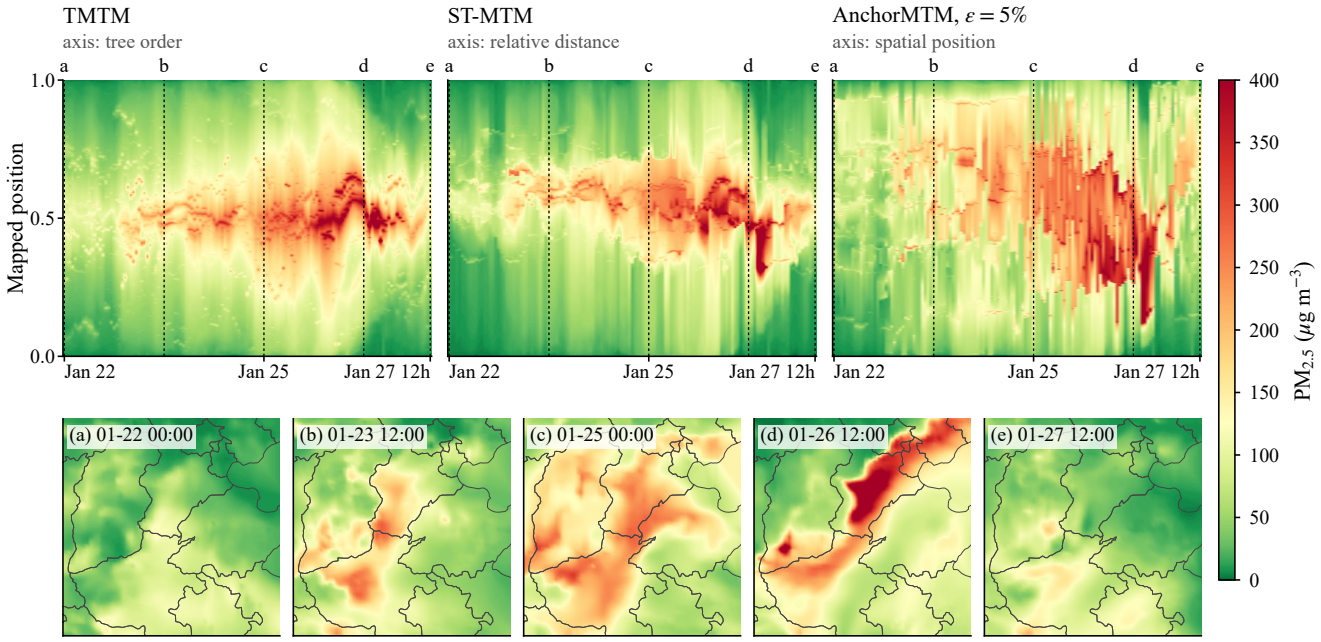


Figure S4: Hourly $PM_{2.5}$, 22 January 00:00 to 27 January 12:00, 2017 (Beijing time, 133 hours), drawn by TMTM, ST-MTM and AnchorMTM ($\varepsilon = 5\%$) on one color scale. The vertical axes differ in meaning, as stated under each title. Dashed lines a–e mark the five snapshots below (110–120°E, 32–41°N, with province borders). Concentrations above 400 $\mu g/m^3$ saturate.